Tutorial-05

(Q.I) Find the regular singular points of the following differential equations:

1. $2x^2y''+3xy'+(x^2-4)y=0$. Ans: x=0
2. $x^2(x-2)^2y''+2(x-2)y'+(x+3)y=0$. Ans: x=2
3. $9x(1-x)y''-12y'+4y=0$. Ans: x=0
4. $(x+1)y''-3xy'+2y=0$. Ans: x= -1

(Q.II) Solve the following differential equations by power series method:

5. $y''+x^2y=0$ Ans: $y=a_0\left(1-\frac{x^4}{12}+\frac{x^8}{12\times7\times8}-\frac{x^{12}}{12\times7\times8\times11\times12}+\cdots\right)+a_1\left(x-\frac{x^5}{20}+\frac{x^9}{20\times8\times9}\cdots\right)$
6. $\left(1-x^2\right)y''-2xy'+2y=0$ about x=0. Ans: $y=a_0\left(1-x^2-\frac{x^4}{3}-\frac{x^5}{5}+\cdots\right)+a_1x$
7. $\left(1+x^2\right)y''+xy'-y=0$ about x=0. Ans: $y=a_0\left(1+\frac{x^2}{2}-\frac{x^4}{8}+\cdots\right)+a_1x$
8. $y''-2xy'+y=0$. Ans: $y=a_0\left(1-\frac{x^2}{2!}-\frac{3x^4}{4!}-\frac{3\cdot7x^6}{6!}-\cdots\right)+a_1\left(x+\frac{1}{3!}x^3+\frac{1\cdot5}{5!}x^5+\frac{1\cdot5\cdot9}{7!}x^7+\cdots\right)$

(Q.III) Solve the following differential equations by Frobenius method:

9. $x(x-1)y''+(3x-1)y'+y=0$ Ans: $y=A\left(1+x+x^2+x^3+\cdots\right)+B\log x\left(1+x+x^2+x^3+\cdots\right)$
10. $3xy''+2y'+y=0$. Ans: $y=a_0\left(1-\frac{x}{2}+\frac{x^2}{20}-\frac{x^3}{480}+\cdots\right)+b_0x^{1/3}\left(1-\frac{x}{4}+\frac{x^2}{56}-\frac{x^3}{1680}+\cdots\right)$
11. $xy''+y'+x^2y=0$. Ans:
$$y=(c_1+c_2\log x)a_0\left(1-\frac{x^3}{3^2}+\frac{x^6}{3^2\cdot6^2}-\frac{x^9}{3^2\cdot6^2\cdot9^2}+\cdots\right)$$
$$+2c_2a_0\left(\frac{x^3}{3^3}-\frac{x^6}{3^5\cdot2^2}\left(1+\frac{1}{2}\right)+\frac{2x^9}{3^9\cdot2^2}\left(1+\frac{1}{2}+\frac{1}{3}\right)+\cdots\right)$$
12. $x(1-x)y''-(1+3x)y'-y=0.$

Ans:
$$y=(C_1+C_2\log x)\left(1\cdot2x^2+2\cdot3x^3+3\cdot4x^4+\cdots\right)$$
$$+C_2\left(-1+x+5x^2+11x^3+\cdots\right)\ where\ C_1=\frac{c_1a_0}{2},\ C_2=-b_0c_2$$
